Tidal potential measured with distant optical clocks

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The Moon and the Sun alter Earth's gravitational potential through their tidal fields and the responses of the solid Earth and oceans. Changes in the tidal potential modify the relative ticking rates of clocks at different locations. Advances in optical clocks and fiber-optic network technologies have made clocks precise probes of spacetime, providing an experimental foundation for the long-term vision of monitoring Earth's gravitational potential with a global clock network. In this work, we use strontium and ytterbium clocks separated by 692 km to detect tidal variations in the potential difference between the coastal region of Shanghai and the inland region of Wuhan. Based on 545.9 hours of observations spanning 353 days, the joint measurement of solid-Earth and ocean loading agrees with model predictions with a consistency of 0.96 ± 0.11. Using spectral analysis, we separate the principal lunar semidiurnal tidal constituent M_2 and the principal solar semidiurnal tidal constituent S_2, with measured equivalent-height power spectral densities of 19.80 and 7.07 cm²/μHz at 22.361 and 23.147 μHz, respectively. This experiment demonstrates that optical clock networks can already be used for high-precision gravitational potential calibration and detection, laying a foundation for related applications.

Introduction

In the weak-field limit of general relativity, a potential difference ΔU produces a fractional frequency shift $\Delta\nu/\nu \simeq \Delta U/c^2$, where c is the speed of light.[1] Optical clocks have become precise probes of spacetime, and their evaluated fractional systematic uncertainties now approach 10^{-19}.[2] Fibre networks and free-space frequency transfer now connect distant optical clocks,[3–5] while proposed quantum clock networks[6] have

advanced to an elementary demonstration using entangled clocks.[7] Near Earth's surface, fractional shifts of 10^{-18}, 10^{-19} and 10^{-20} correspond to equivalent heights of approximately 1 cm, 1 mm and 0.1 mm, respectively.[8,9] Clock measurements complement conventional levelling and satellite gravity observations,[10,11] and have resolved gravitational redshift within a millimetre-sized atomic sample.[9] Precision clock comparisons also support searches for ultralight dark matter and the development of future gravitational-wave detectors.[12,13]

Earth's dynamic environment offers both signals to measure and backgrounds to understand. Mass redistribution and deformation accompany volcanic activity,[14] ocean and atmospheric disturbances generate tsunamis,[15] and changing water storage records climate variability.[16] Monitoring these processes is important for understanding the environment and its hazards. Their contributions to gravitational potential also overlap in a clock record, so a signal of interest in one measurement can become a background in another. Extracting weaker signals therefore requires identifying the sources of measured variations and assessing how accurately each can be removed. Tides provide a predictable signal with which to develop this capability. The Moon and Sun alter the potential directly and through deformation of the solid Earth and movement of ocean water. Their contributions must be distinguished from one another and from laboratory variations synchronized to the solar day. Observational gaps further distribute spectral power into neighbouring frequencies. Separating the tidal components and their sampling sidelobes is therefore important for subtracting the tidal background and identifying weaker signals in the residual record.

Here we measure tidal changes in the potential difference between a strontium clock in coastal Shanghai and a ytterbium clock in inland Wuhan, separated by 692 km and connected by a 1,350-km fibre link. From 545.9 hours of observations spanning 353 days, we measure the combined solid-Earth and ocean-loading tide and separately detect the principal lunar and solar-frequency semidiurnal contributions. The two contributions evolve differently in phase and produce neighbouring spectral peaks. Each remains identifiable after the other contribution is removed. These measurements test how individual tidal contributions can be identified and removed before weaker potential variations are sought.

Results

Measuring a coastal–inland potential difference

The Shanghai Sr clock and Wuhan Yb clock measure tidal changes in potential difference along a 692-km baseline (Fig. 1a). A 1,350-km phase-stabilized fibre link connects them through Hefei.[17] The comparison instability is $5.4 \times 10^{-16}/\sqrt{\tau/\mathrm{s}}$, where τ is the averaging time, and is dominated by the clocks. Changes in their frequency ratio track the changing potential; a separate frequency offset for each observing segment absorbs the static contribution.

The coastal–inland geometry gives ocean loading an appreciable contribution to the predicted signal. The solid-Earth calculation follows the IERS conventions,[18] and the

FES2022 loading calculation[19,20] includes water attraction, the induced Earth potential and clock displacement.[21] For the principal lunar semidiurnal tide, M_2, the differential solid-tide and loading amplitudes are 3.41×10^{-18} and 1.42×10^{-18} in fractional frequency. Their 20° phase difference gives a combined amplitude of 4.77×10^{-18} (Fig. 1b).

The record spans 353 days across five campaigns and 54 observing segments (Fig. 1c). Figure 3 uses 3,299 non-overlapping 600-s means containing 545.924 h of retained one-second samples. The triangular-filter reduction applies additional transient screening and a 1,200-s filter at 600-s steps, retaining approximately 544 h. Each reduction retains its own weights and physical-template realization, with segment offsets in both (Methods). The 274-h summer record overlaps the earlier absolute frequency-ratio study; Table 1 compares estimates from those shared observations.[17]

Tidal amplitude, phase and observation coverage

The non-overlapping reduction gives a full-template response of 0.96 ± 0.11, consistent with the combined potential prediction (Fig. 3a). A response of one denotes agreement with the predicted change in frequency ratio. To distinguish the semidiurnal contributions, we group M_2 with the larger lunar elliptic semidiurnal tide, N_2, which reflects the Moon's elliptical orbit. A second group combines the principal solar semidiurnal tide, S_2, with the lunisolar semidiurnal tide, K_2, associated with lunar and solar declination. Jointly fitting these groups gives responses of 1.00 ± 0.13 and 0.93 ± 0.36. Their internal phases and amplitude ratios follow the physical model. The uncertainties are one statistical standard error, using the larger of scaled least-squares and UTC-day-deletion jackknife estimates. Allowing a lunar term shifted by a quarter cycle gives in-phase and quadrature responses of 1.00 ± 0.13 and -0.10 ± 0.20 (Fig. 3b), consistent with the predicted timing.

The triangular-filter reduction gives lunar and neighbouring semidiurnal responses of 0.97 ± 0.14 and 0.90 ± 0.29, with coefficient correlation -0.32 (Fig. 2 and Extended Data Fig. 11a,b). Residual resampling calibrates the lunar statistical uncertainty. A free-phase lunar fit gives amplitude 0.98 ± 0.11 and phase offset $-6° \pm 8°$, with segment-bootstrap uncertainties. Both reductions therefore support the predicted amplitude and lunar phase, while retaining their distinct sampling and error definitions. The non-overlapping amplitude fit weights observations by the noise in each observing period. The triangular reduction applies the same filter to data and model, and its baseline grouped fit includes no empirical daily terms (Methods).

The summer observations distinguish the components by covering their changing relative phase. The lunar tide repeats about every 12.42 h, whereas the solar tide repeats every 12 h. Their relative phase traverses a full cycle during the two-month summer campaign; shorter campaigns cover narrower intervals (Fig. 1d). Summer supplies 61% of the conditional information on the lunar response in the triangular-filter fit and gives 0.87 ± 0.18, compared with 1.05 ± 0.31 for the earlier campaigns together. The five estimates are consistent ($\chi^2 = 4.1$, four degrees of freedom, $p = 0.40$; Extended

Data Table 3). Figure 2 shows the summer time series and lunar phase dependence. Its axis compresses gaps, whereas Extended Data Fig. 4 retains calendar time. The triangular-filter full-template and semidiurnal responses are 0.90 ± 0.12 and 0.96 ± 0.12; their amplitude, covariance and full-record spectral checks are presented together in Methods and Extended Data Fig. 11. Semidiurnal estimates agree across campaigns ($\chi^2 = 3.2$, four degrees of freedom, $p = 0.52$; Extended Data Fig. 3). Subtracting the fitted full template reduces lunar spectral power by 99% in this reduction (Extended Data Fig. 11c).

Spectral density and separate semidiurnal detections

The equivalent-height power spectral density has lunar and solar-frequency peaks at 22.361 and 23.147 μHz, with densities of 19.80 and 7.07 cm²/μHz (Fig. 3c). These are scan maxima near the prescribed astronomical frequencies. The scan uses the non-overlapping observations, equal weights and projection against the 54 segment constants. Both peaks decrease after subtracting the fixed unit-amplitude physical prediction. Identical normalization and linear axes allow the observed and residual spectra to be compared directly; Methods defines the density and equivalent-height conversion.

Each principal frequency retains structure when the other is controlled with a freely phased sinusoid (Extended Data Fig. 7). The conditional scans also control the smaller neighbouring N_2 and K_2 contributions. Joint sinusoidal fitting gives M_2 and S_2 fractional-frequency amplitudes of 4.50×10^{-18} and 1.92×10^{-18}, with coefficient uncertainties in Extended Data Table 4. These single-frequency estimates allow independent phases, whereas the grouped responses scale predictions with prescribed internal phase and amplitude relationships. Together, the local and conditional spectra identify two contributions without imposing their astronomical phase relation.

Reliability of the separated signals

Reciprocal subtraction quantifies how much each component changes when the other is removed (Extended Data Fig. 8). Subtracting the jointly fitted lunar waveform changes the single-frequency solar-frequency amplitude from 2.918×10^{-18} to 1.921×10^{-18}. Subtracting the solar-frequency waveform changes the lunar amplitude from 4.857×10^{-18} to 4.505×10^{-18}. Both remain identifiable. The maximum canonical correlation, which measures the strongest overlap between linear combinations of their projected sine–cosine pairs, is 0.290; the largest absolute cross-component coefficient correlations are 0.248 under scaled least squares and 0.261 under UTC-day deletion (Extended Data Fig. 9). These results measure the coupling induced by the sampling and support separate detection. Methods gives the conditional scans, compensation curves and correlation calculations.

The triangular-filter lunar estimate also remains consistent under explicit daily-disturbance models (Fig. 4). A 10^{-18} fractional-frequency perturbation at 24 h or 12 h changes it by at most 0.020 when its phase is fixed across the record. Allowing independent campaign phases increases these bounds to 0.032 and 0.12, respectively.

Fits with common 24-h and 12-h terms give 1.00 ± 0.13; campaign-specific free-phase 12-h terms give 0.82 ± 0.17, or 0.86 ± 0.18 with 24-h terms included. Both clocks correct blackbody shifts in real time. Temperature regressions and the chamber-temperature record provide further checks (Methods and Extended Data Fig. 2).

Instrumental and processing checks give small changes in the lunar response. Adding the fibre-link monitor on matching clock samples changes the lunar and full-template coefficients by less than 0.005. Retaining the excluded transients gives 0.96 ± 0.14, compared with 0.97 ± 0.14 after screening. Group resampling gives a 95% lunar interval of [0.76, 1.18]. Fits also predict withheld observations: the full template improves seven of eight periods, and the lunar component improves six, with pooled residual-sum-of-squares reductions of 4.9% and 1.7%. Selection, alternative offsets and processing checks are detailed in Methods and Table 1. Coastal-model sensitivity is larger: extending the ocean mask near Shanghai by 0.3° or 0.6° changes the response by about −0.06 or −0.07. These model shifts are reported separately from statistical uncertainty.

Ocean loading and diurnal response

The triangular-filter observations constrain ocean loading and the once-daily tidal band less precisely than the principal semidiurnal response. The separately fitted ocean-loading coefficient is 0.89 ± 0.55 (Methods). A two-band fit, estimating diurnal and semidiurnal responses together, gives a diurnal coefficient of 0.36 ± 0.50 (Extended Data Fig. 3b). Its error is the larger of the scaled least-squares and day-by-day jackknife errors. This coefficient describes the physical diurnal tidal prediction; the empirical 24-h terms above test solar-day-synchronous disturbances. The combined potential agrees with the observations, while these separate estimates leave broader uncertainty in how its contributions are apportioned. Refining the coastal loading representation would also sharpen the model comparison.[22]

Discussion

The observations resolve two neighbouring semidiurnal contributions despite gaps in operation. Those gaps also produce spectral sidelobes that matter when interpreting weaker peaks. Passing the lunar waveform through the measured sampling reproduces neighbouring features at 22.1016 and 22.6283 μHz (Extended Data Fig. 10). Their powers fall by 99.56% and 99.39% after lunar subtraction, with minima at 0.996 and 1.0445 times the jointly fitted waveform. This behaviour identifies a substantial sampling-leakage contribution at those frequencies. Other peaks respond differently. The total time span sets the scale of frequency resolution, while the distribution of observing intervals determines sidelobes and component coupling. Thus, subtracting an identified component helps distinguish its sampling response from residual structure. At the solar semidiurnal frequency, physical attribution additionally requires environmental information because a laboratory 12-h disturbance occupies the same frequency.

Comparison stability and phase coverage together determine the precision of individual components. The observed semidiurnal response is constrained by its accumulated

phase evolution across campaigns. Separating ocean loading from the solid-Earth contribution is harder because their waveforms overlap; daily tidal terms also compete with segment offsets and solar-day variations. The reported coefficient errors express these differences on the available record. Existing residual-injection tests recover the lunar response under the tested daily disturbances, while alternative-processing simulations examine interval coverage under several temporal-noise models (Methods). Under white-frequency-noise averaging, improved stability reduces the time needed to reach a chosen signal level. An average must still preserve the signal's time dependence, and observations must sample its cycles and phases. Stability alone therefore does not specify the precision achievable for a given component.

The present clocks and fibre link already support observation of daily tidal-potential variations. Sustained network operation is the next experimental step. Longer, more continuous records would improve phase coverage and reduce ambiguity from the sampling window; better clock stability would permit weaker changes to be resolved over shorter intervals. Each endpoint needs stable operation and records of environmental and local potential changes. Additional sites would provide spatial information to help identify sources. These developments extend demonstrated sub-centimetre-equivalent clock performance[8] towards measurements of weaker geophysical variations after identifiable tidal contributions are removed. Precision comparisons also support ultralight-dark-matter searches[12] and the development of future clock-based gravitational-wave detectors.[13]

Summary

We measured tidal potential differences between Shanghai and Wuhan optical clocks and separately detected lunar and solar-frequency semidiurnal contributions with equivalent-height spectral densities of 19.80 and 7.07 cm²/μHz. Amplitude and phase estimates, conditional spectra, reciprocal subtraction and correlation analysis support the separated signals. Their identification also allows specific sampling sidelobes to be traced to their source. Sustained observations at more sites would extend this capability to weaker potential changes and their spatial structure.

Note added. A related observation of tidal potential variations in a European optical-clock network was reported during preparation of this work.[23]

Methods

Clocks, frequency transfer and sampling

The Sr optical lattice clock is located in Shanghai (31.130° N, 121.570° E, height approximately 5 m) and the Yb optical lattice clock in Wuhan (30.529° N, 114.357° E, approximately 30 m). The tidal model is insensitive to the absolute heights: changing the Shanghai height by 5 m changes the differential model by 4×10^{-23}. The geopotential difference between the clocks, 280.042 ± 0.085 m^2 s^{-2}, was determined by first-order levelling [17]. It enters the static frequency ratio and is absorbed by the fitted offsets. The clocks, fibre link and frequency chain are described in ref. [17]. The evaluated systematic

uncertainties of the Sr and Yb clocks are 9.2×10^{-19} and 1.3×10^{-18}, respectively [24,25]; optical frequency dissemination over field fibres of comparable length has also been demonstrated [26,27].

A 1,550-nm carrier referenced to the Sr clock through a frequency comb in Shanghai is transmitted to Wuhan, where its beat against the Yb-referenced comb is recorded at 1-s intervals. In the 2026 campaigns, the beat frequency is 33.6 MHz and a deviation of 1 Hz corresponds to 5.17×10^{-15} in Yb/Sr. In 2025, the counter recorded the beat after division by two; these deviations are therefore multiplied by two. Fitting the September–October 2025 campaign alone gives $A = 1.28 \pm 0.31$, consistent with this scale and sign. The white-frequency-noise level of the tide-corrected ratio, $\sigma_y\sqrt{\tau}$, obtained from the overlapping Allan deviation [28] at averaging times of 256–4,096 s, is 7.9×10^{-16} in September–October 2025 and 4.2–5.3×10^{-16} in the 2026 campaigns.

The record contains five campaigns: September–October 2025, January 2026, April 2026, May 2026 and June–August 2026. The 54 segments were defined by the experiment team; the summer segments are those used in ref. [17]. There are 38 segments and 270 h before June and 16 segments and 274 h in June–August, giving 544 h, or approximately 23 days, of retained data. The analysis distinguishes these five campaigns from eight observation periods and 29 operating groups. Campaigns define the seasonal subsets and campaign-specific daily terms; observation periods are the units of held-out prediction and period-level sign resampling; operating groups are the units of group bootstrap resampling; segments carry the offsets in the primary fit. These levels are not treated as interchangeable independent samples.

Data selection

The same selection procedure is applied to every segment. The time axis of each data file is reconstructed from its first time stamp and the sample index. Samples more than 10 Hz from the segment median are removed, the longest uninterrupted run is retained, and leading and trailing samples deviating from the median by more than 1% of the peak-to-peak range are trimmed. The 10-Hz threshold corresponds to approximately 5×10^{-14} in Yb/Sr. To identify smaller transient excursions, the retained 1-s data are averaged over 300 s and compared with a running median over ±30 min. Averages deviating by more than six robust standard deviations of these differences within the segment are removed together with their neighbours. This step removes 0.4% of the samples, or 2.2 h, including an excursion of approximately -2×10^{-16} lasting about 20 min on 7 July 2026.

The transient criterion uses no tidal model. Adding the model with factors of -1, 1 or 3 leaves the removed samples unchanged. Six robust standard deviations is the lowest tested threshold for which this invariance holds up to three times the model. For block lengths of 60 or 300 s and thresholds from four to eight standard deviations, the lunar amplitude ranges from 0.93 to 1.02. Retaining the excursions gives 0.96 ± 0.14; the full-template and semidiurnal amplitudes change by less than 0.04. Before transient removal, the procedure reproduces the summer sample set of ref. [17]. After removal,

the campaign diurnal amplitudes are mutually consistent ($p = 0.21$). Because the tested tidal additions do not change selection and the subsequent filter is linear, the injection simulations use the selected data.

Durations refer to retained 1-s samples; exposure measured from the filtered data at 600-s steps, used for temperature overlap, is slightly shorter. The last April segment ends at 18:00 on 19 April, following the experiment team's segment table. Omitting endpoint trimming while retaining the other steps changes the lunar amplitude by −0.009 and the full-template amplitude by 0.008. Each segment has its own fitted offset. Time stamps are in local time (UTC+8); shifting the model by ±10 min changes A by less than 0.02. Long-period tides are outside the analysis: the similar site latitudes make the predicted differential semiannual tide only about 2×10^{-20}.

Solid Earth tide and ocean loading

The solid Earth contribution at each clock is computed directly as

$$\delta W = \sum_m (1 + k_{2m}) V_{2m} + (1 + k_3) V_3 - g u_r.$$

Here V_{2m} are the degree-2 tide-generating potentials of the Moon and Sun separated by order m, V_3 is the lunar degree-3 potential, and u_r is radial displacement. We use $k_{20} = 0.29525$, $k_{22} = 0.29801$ and $k_3 = 0.093$. The IERS 2010 resonance formula is applied constituent by constituent for k_{21}. It gives 0.2588 for the lunisolar diurnal tide, K_1, 0.2873 for the principal solar diurnal tide, P_1, and 0.2975 for the principal lunar diurnal tide, O_1. These diurnal constituents describe approximately once-daily changes associated with lunar and solar declination. The radial displacement is computed with the IERS algorithm, including the frequency and latitude dependence of the Love number h [18].

The ocean-loading potential is computed by grid convolution:

$$\delta W_{\text{ocean}}(t) = \sum_{\text{ocean}} \rho_w \zeta(\mathbf{r}', t) G_W(\psi)\, \mathrm{d}A, \qquad G_W(\psi) = \frac{ga}{M_E} \sum_n (1 + k'_n - h'_n) P_n(\cos\psi).$$

The ocean elevation ζ is taken from FES2022 on a $1/30^\circ$ grid [19,20], $\rho_w = 1,025\ \mathrm{kg\,m^{-3}}$, a and M_E are Earth's radius and mass, and ψ is angular distance. The Green's function follows ref. [21]. PREM load Love numbers are used through degree $n = 696$ [29], from the gravity-toolkit distribution [30]. Above this degree, h'_n is joined smoothly to its Boussinesq limit of −6.22, and the series is summed with a Kummer transformation through $n = 40,000$. The convolution uses full grid resolution within 3° of each station and sums blocks of 8×8 cells farther away. This aggregation changes differential M_2 and K_1 loading by at most 0.05%. The ocean mean is removed to enforce mass conservation. The larger lunar elliptic diurnal tide, Q_1, accounts for a further once-daily contribution associated with the Moon's elliptical orbit. These four diurnal constituents and the

four semidiurnal constituents introduced above (N_2, M_2, S_2 and K_2) are synthesized with astronomical arguments and nodal corrections.

The full loading potential exceeds the vertical-displacement contribution $-g\delta h$ by factors of approximately 1.4 for the semidiurnal constituents and 1.8 for the diurnal constituents. The radial displacement from the same convolution agrees with the FES2022 load-tide grid to within 2% in amplitude and 1.2° in phase for seven of the eight constituents. This comparison checks the convolution and displacement contribution; the direct attraction and induced-potential terms use the same PREM load Love numbers. For S_2, the amplitude differs by 18% and the phase at Wuhan by 54°; the load-tide grid uses a different ocean-model release. The S_2 loading contributions expressed as fractional frequency are 0.62×10^{-18} in Shanghai and 0.18×10^{-18} in Wuhan, giving a differential amplitude of 0.63×10^{-18}, compared with 1.42×10^{-18} for M_2. Substituting the S_2 loading inferred from the load-tide grid changes the full-template amplitude by 0.001 and the semidiurnal amplitude by 0.000 at the reported precision, with no reported change in the lunar amplitude. The lunar and near-12-h groups have separate fitted amplitudes.

The Shanghai clock is 22 km from the nearest ocean cell. To examine coastal-mask sensitivity, land cells within 0.3° or 0.6° of the coastline in a $4^\circ \times 5^\circ$ box centred on Shanghai are filled with the nearest ocean values. These strips, approximately 33–66 km wide, increase the Shanghai M_2 loading potential expressed as $\delta W/g$ from 15.3 mm to 17.0 or 18.5 mm and decrease the fitted response by 0.06 or 0.07. These are physical-model sensitivity tests, with shifts approximately half the quoted lunar statistical uncertainty; they are not treated as independent Gaussian errors or as an exhaustive loading-model uncertainty.

Tidal fits and uncertainty

The 1-s observations and model are filtered identically using a 1,200-s triangular window stepped by 600 s. The whole-template model is fitted by least squares as $y(t) = c_i + As(t)$, with an offset c_i for segment i. For the band analysis, the eight-constituent model is divided either into semidiurnal (M_2, S_2, N_2, K_2) and diurnal (K_1, O_1, P_1, Q_1) bands, or into lunar (M_2, N_2), near-12-h (S_2, K_2) and diurnal groups with separate amplitudes. The primary grouped fit contains no empirical daily terms. If the full physical model is correct, fitting the solid Earth tide alone on the actual sampling is expected to give $A = 1.32$, because loading adds to the solid tide with an M_2 phase difference of approximately 20°.

Where indicated, uncertainty is the larger of the scaled least-squares standard error[31] and the day-by-day jackknife error. For the full template, semidiurnal band and lunar group, uncertainty is calibrated by residual resampling. The fitted residuals are shifted circularly within each continuous run of the filtered record and added to the model with $A = 1$; the standard deviation of the refitted amplitude over 200 realizations supplies the calibrated error. These shifts retain the within-run residual structure under circular wrapping. For the lunar group, two independent sets of 200 realizations give

standard deviations of 0.13 and 0.14, compared with 0.13 for the larger of the scaled least-squares and jackknife errors. We quote 0.14; the resampling bias is +0.01. With the daily disturbances described below, 5,000 realizations per scenario give standard deviations rounding to 0.14, coverage of 0.67–0.68 for the interval ±0.14, and 0.945–0.952 for ±1.96 × 0.14.

Calibrated errors are used for the whole-template and semidiurnal responses of individual campaigns and of the full record. For the diurnal and near-12-h groups, fits containing empirical daily terms, and fitted daily amplitudes, the larger of the scaled least-squares and jackknife errors is used. In the resampled data, these errors differ from the calibrated uncertainty by factors of 0.8–1.3. Unless otherwise specified, quoted ± uncertainties denote one standard deviation or one standard error as defined by the applicable procedure.

Group-bootstrap intervals are obtained by drawing the 29 operating groups with replacement and refitting 1,000 times; the segment bootstrap instead draws the 54 segments. These procedures permit correlations within each resampling unit but assume independence between units. For the semidiurnal response, the group-bootstrap standard deviation of 0.08 is smaller than the calibrated value of 0.12, and the larger value is quoted. In the phase test, each tidal band has an in-phase and a quadrature template, the latter formed by shifting every constituent by 90°. Segment-bootstrap standard deviations give a lunar amplitude of 0.98 ± 0.11 and phase of $-6^\circ \pm 8^\circ$.

The phase-folded summer display is formed after subtracting the fitted segment offsets and non-lunar groups and then averaging in 30° M_2-phase bins; the M_2+N_2 model is folded with the same sampling. Its day-by-day jackknife error bars hold the fitted non-lunar terms fixed. They are conditional display uncertainties and do not propagate uncertainty in those fitted terms; they are distinct from the uncertainty of the joint lunar-amplitude estimate. The summer time-series display uses 3-h averages and removes gaps longer than 6 h from its horizontal axis, marking their locations.

The eight constituents explain 99.84% of the model variance. The remaining component, with an r.m.s. of 0.18×10^{-18}, is retained at its predicted value. Setting it to zero or doubling it gives semidiurnal amplitudes of 0.95–0.96 and lunar amplitudes of 0.95–0.99. An additional central-value check uses a Kalman filter with state (segment offset, A), resetting the offset at each segment and allowing a random walk reaching 5×10^{-18} after 6 h. It gives $A = 0.95$ for the full record; changing that scale to 2×10^{-18} or 10^{-17} gives 0.92 or 0.99, respectively.

Daily disturbances and validation

Daily sensitivity fits add terms $a_1 \cos \omega t + b_1 \sin \omega t + a_2 \cos 2\omega t + b_2 \sin 2\omega t$, where $\omega = 2\pi/(1\ \text{day})$ and t is local time. Common coefficients and campaign-specific coefficients are examined. A physically constrained alternative uses a free-amplitude 24-h term in each campaign with the phase of the 24-h harmonic of hourly ERA5 2-m air temperature at the two sites [32], averaged over that campaign. Another alternative includes the air-temperature anomalies at both sites relative to a 24-h running mean as two re-

gressors. Within a short campaign, a 24-h term is partly degenerate with diurnal tides and segment offsets.

The sensitivity of the lunar estimate to $B\cos(\omega t-\varphi)$ is evaluated using the lunar row h of the least-squares operator. For a common phase, the maximum amplitude change is

$$B\sqrt{(h\cdot\cos\omega t)^2+(h\cdot\sin\omega t)^2}.$$

For phases independent between campaigns, the maximum is the sum of the corresponding campaign norms. These bounds describe the specified sinusoidal disturbances and phase assumptions; they do not exclude arbitrary time-varying daily effects.

Injection tests use circularly shifted residuals of the full-template fit, adding either the full tidal model ($A = 1$) or the model without its lunar group ($A = 0$). Four daily-disturbance scenarios are considered: no added daily disturbance; campaign-specific 24-h terms with air-temperature phases and amplitudes fitted to the observations; common 24-h and 12-h terms with fitted amplitudes and phases; and the fitted summer 24-h and 12-h terms. The added disturbances have r.m.s. values up to 1.9×10^{-18}. In 300 realizations per scenario and injected amplitude, lunar-amplitude biases lie between -0.03 and -0.00 at the reported precision. The larger of the scaled least-squares and jackknife errors covers the injected value in 0.59–0.71 of realizations at one standard error and 0.90–0.95 at 1.96 standard errors. The main estimate uses the calibrated uncertainty above. For $A = 1$, the fitted lunar amplitude exceeds three times its estimated error in every realization; for $A = 0$, this fraction is at most 0.003. These simulation results are conditional on the selected residual and disturbance models.

A complementary restricted wild bootstrap flips the residual sign of whole observation periods under the null. With eight periods, the smallest attainable two-sided probability is $2/256$. The resulting two-sided probabilities are 0.008 for the lunar estimate and 0.008 for the whole-template estimate. Campaign information shares are calculated by projecting the lunar template onto the complement of the other fit columns—near-12-h and diurnal templates and segment offsets—and summing its squared projected values within each campaign.

Held-out prediction estimates the amplitude from the other observation periods and applies it to the excluded period, fitting that period's segment offsets. The whole-template prediction reduces residual sums of squares in seven of eight periods, by 4.9% when pooled; the lunar prediction improves six of eight periods, by 1.7% when pooled. These checks use the same underlying record as the primary fits.

Spectral diagnostics and alternative processing

The full-record spectrum shows sinusoidal power after projection of the segment offsets, before and after subtraction of the fitted whole template. Its 99% threshold is the quantile of the maximum over 5–40 µHz in 200 resampled residual series. For the April record, sinusoidal power is expressed as amplitude squared divided by two. Pointwise 99% thresholds and the 99% threshold of the maximum over 15–50 µHz are obtained

from 1,000 resampled residual series. No frequency exceeds the latter threshold after tidal correction; the largest residual feature, near 39 µHz, is away from the tidal frequencies. The four-day April record combines neighbouring constituents into broad diurnal and semidiurnal bands. Spectral displays use a linear vertical scale. A decrease in spectral power after fitted-template subtraction is a consistency check, not an independent detection statistic.

As a processing cross-check, the segment-wise estimator of ref. [17] demeans filtered data and model within each segment, fits the amplitude through the origin using non-overlapping 1,200-s windows, and combines segment estimates with inverse-variance weights. It differs from the primary joint fit in both sampling and weighting. For the summer record it gives $A = 0.78 \pm 0.19$ with the present physical model, compared with 0.87 ± 0.15 with the model of ref. [17]; the primary joint lunar fit gives 0.87 ± 0.18. The lower summer whole-template response, 0.67 ± 0.18, accompanies a low diurnal response and should not be substituted for the lunar estimate. For April, the segment-wise and joint estimates are 0.66 ± 0.31 and 0.91. For the complete record, the segment-wise result is 0.84 ± 0.12, with uncertainty scaled by the square root of the segment-to-segment reduced chi-square of 2.1.

Independent processing of the same 2026 records uses non-overlapping 600-s means, corrected clock systematic shifts and fixed ten-day calendar baselines. It gives a whole-template response of 0.88 ± 0.10 without empirical daily terms and a lunar response of 0.89 ± 0.28 with daily terms in each ten-day block. Its whole-template prediction improves all six held-out 2026 periods. Simulations with white, Ornstein–Uhlenbeck (correlation times of 1 h to 3 d), Student-t and moving-average noise give biases below 0.04 and nominal 95% interval coverage of 94.5–99.8%, using the largest of least-squares, delete-day and delete-period standard errors. Simpler solid-tide approximations change its whole-template scale by less than 0.01. These estimates are processing checks on shared observations and are not combined as independent measurements.

Fibre-link monitor

The round-trip beat of the fibre link is recorded at the Shanghai end near 10 MHz with 1-s resolution. Its deviations are normalized by the optical carrier frequency without division by two and filtered in the same way as the clock ratio. Samples more than 10 Hz from the median are excluded. Monitor excursions reaching 10^{-12} leave the recorded ratio unchanged. Additionally excluding 300-s monitor blocks deviating by more than 5×10^{-16} changes the regression result by less than 0.005. The monitor covers 474 h of the analysed data. On that overlap, the lunar and whole-template amplitudes are 0.93 and 0.80; including the monitor as an additional regressor changes each by less than 0.005.

Environmental analysis

The Sr vacuum-chamber temperature is the mean of 20 PT100 sensors sampled every 20 s, available from 28 June to 7 July 2026 and overlapping 98 h of the filtered summer

record. The blackbody sensitivity is 6.26×10^{-17} K^{-1} at 294.3 K; a temperature increase raises Yb/Sr. The daily temperature profile is obtained after subtracting a 24-h running mean and excluding the chamber-temperature transient on 30 June. Both clocks apply blackbody corrections in real time, so the recorded ratio already includes them. The regression therefore tests correction errors proportional to chamber temperature; its expected coefficient is zero.

Over the 98-h overlap, the blackbody shift calculated from chamber temperature and filtered like the ratio has an r.m.s. of 2.8×10^{-18}, including the 30 June transient. Adding it with a free coefficient gives 0.92 ± 0.74, or 0.72 ± 0.74 after excluding the transient, with day-by-day jackknife errors. Both are consistent with zero at this precision. Through the summer lunar-estimator response of 0.016 per 10^{-18}, a residual with the size of the observed 24-h ratio term would change the lunar amplitude by about 0.045. Without real-time correction, the blackbody shift would contain a daily component of 2.0×10^{-18} peaking at 12:13 local time. In the same overlap, the fitted 24-h ratio term is $(0.9 \pm 1.1) \times 10^{-18}$ with maximum at 10:54 ± 5.0 h. The mean daily profile of the uncorrected blackbody contribution has a peak-to-peak range of 4×10^{-18}, indicating the size of the removed effect.

Additional model and disturbance checks

On the primary sampling after segment-mean removal, the predicted solid-tide and loading root-mean-square fractional frequencies are 3.0×10^{-18} and 1.35×10^{-18}, and their sum is 4.1×10^{-18}. The whole-template response fitted to the solid-Earth tide alone is 1.19 ± 0.20, compared with 1.32 expected on this sampling for the combined model. The separately fitted loading coefficient is 0.89 ± 0.55, consistent with the combined prediction but insufficient to establish a separate loading detection.

The summer 24-h term has amplitude $(2.8 \pm 1.3) \times 10^{-18}$ and peaks at 15:48 local time, with a 1.5-h standard uncertainty; the 12-h term is $(0.6 \pm 0.7) \times 10^{-18}$. Their summer projection coefficients are 0.016 and 0.043 per 10^{-18}, giving maximum shifts of 0.045 and 0.024 at those fitted amplitudes (Extended Data Table 3). For the full record, a common-phase 12-h perturbation of amplitude 10^{-18} changes the whole-template response by at most 0.087.

Campaign-specific 24-h regressors with phases fixed by local air temperature give a lunar response of 0.99 ± 0.14; free-phase campaign-specific 24-h terms give 0.97 ± 0.15. Omitting each earlier campaign in turn gives lunar responses between 0.89 and 0.99. Three of the nine identified transient excursions coincide with approximately equal and opposite link-monitor excursions (Extended Data Fig. 5). The transient criterion acts on 300-s deviations from a local running median and does not use the tidal waveform.

Temperature-anomaly regressors at both sites give a lunar response of 0.98 ± 0.13. Replacing segment offsets by operating-group or period offsets gives 0.97 ± 0.11 and 0.99 ± 0.11, respectively. These estimates use the triangular-filter reduction (Fig. 4a).

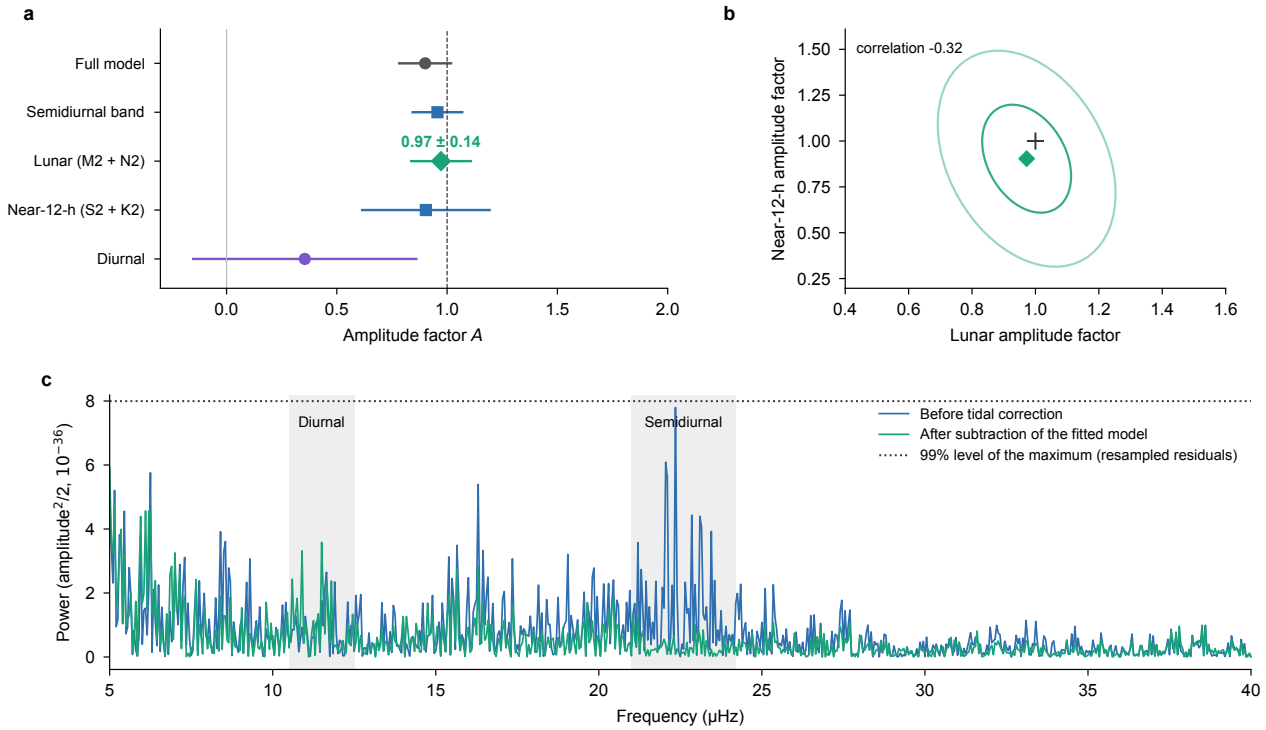
Primary grouped responses and full-record harmonic power

The lunar response is $A_{\mathrm{L}} = 0.97 \pm 0.14$, consistent with the predicted amplitude (Extended Data Fig. 11a). The uncertainty is calibrated by residual resampling and represents one standard deviation (Methods). The neighbouring semidiurnal response is 0.90 ± 0.29, with a correlation of -0.32 with the lunar coefficient (Extended Data Fig. 11b); the diurnal response is 0.35 ± 0.51. A fit allowing an in-phase and a quadrature lunar template gives an amplitude of 0.98 ± 0.11 and phase offset of $-6° \pm 8°$, with segment-bootstrap standard deviations. This phase test and the fixed-phase amplitude estimate describe the same observations.

The full tidal template gives $A = 0.90 \pm 0.12$, and the combined semidiurnal band gives 0.96 ± 0.12. Semidiurnal campaign estimates are consistent ($\chi^2 = 3.2$, four degrees of freedom, $p = 0.52$; Extended Data Fig. 3). The April record gives a whole-template response of 0.91 ± 0.13 over approximately four days (Extended Data Fig. 4).

The full-record spectrum displays semidiurnal structure that is strongly reduced after subtraction of the fitted whole tidal template (Extended Data Fig. 11c). At M_2, harmonic power after projection of segment offsets decreases by 99%. The April spectrum provides a complementary view on an actual calendar-time axis (Extended Data Fig. 4). These plots show squared sinusoidal amplitude divided by two on a linear ordinate, rather than density per unit frequency. Their residual-based thresholds provide frequency-domain evidence alongside the physical-template estimates; the power reduction is not an instrumental signal-to-noise ratio.

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Extended Data Fig. 11 | Primary grouped responses and full-record harmonic power. a, Amplitude factors for the full model and the semidiurnal band, each fitted with one amplitude, and for the lunar (M_2, N_2), near-12-h (S_2, K_2) and diurnal groups fitted jointly. All fits include segment offsets and omit empirical daily terms. Error bars are one standard deviation, calibrated by residual resampling for the full model, semidiurnal band and lunar group; for the near-12-h and diurnal groups, the larger of the scaled least-squares and day-by-day jackknife uncertainties is used. The dashed line denotes the prediction, $A = 1$. **b,** Joint covariance contours at radii one and two for the lunar and near-12-h amplitudes, using the least-squares correlation and scaling the covariance to the quoted marginal uncertainties. These radii do not denote joint 68% and 95% confidence regions. The cross denotes the model prediction. **c,** Sinusoidal power of the full record after projection of segment offsets, before and after subtracting the fitted full model. Power is the squared sinusoidal amplitude divided by two, plotted on a linear vertical axis in units of 10^{-36}; it is not a power spectral density per unit frequency. The dotted line is the 99% quantile of the maximum power over 5–40 μHz from 200 resampled residual series.

Auxiliary grouped-response and phase estimates

Figure 3 uses the non-overlapping reduction, with 3,299 means, 1,965,327 nominal one-second members and 54 segment constants. The 545.924 member-hours span 353.39 days. This reduction retains its existing valid-interval selection and does not apply the primary analysis's triangular filter or additional transient screening. The template realization is also distinct from the direct-potential primary model. Its solid-Earth sequence converts the IERS radial displacements into fractional frequency as

$$y_{\mathrm{solid}} = -\frac{1 + k_2 - h_2}{h_2} \frac{g(u_{\mathrm{Wuhan}} - u_{\mathrm{Shanghai}})}{c^2},$$

with $h_2 = 0.6078$, $k_2 = 0.3010$ and $g = 9.794\,\mathrm{m/s^2}$. Its ocean-loading sequence uses the effective-potential convolution with direct water attraction, induced potential and clock displacement, synthesizing 18 loading constituents. Eight named constituents form the lunar, neighbouring semidiurnal and diurnal groups; the remaining terms form a fixed remainder r_0.

For the complete-template estimate, a single coefficient multiplies the full prediction, $y_i = c_{s(i)} + A_T(s_{L,i} + s_{S,i} + s_{D,i} + r_{0,i}) + \epsilon_i$. For the grouped estimate, the remainder is subtracted once and the three groups are fitted jointly, $y_i - r_{0,i} = c_{s(i)} + A_L s_{L,i} + A_S s_{S,i} + A_D s_{D,i} + \epsilon_i$. The phase-relaxed model adds a quarter-cycle-shifted lunar template while retaining the prescribed internal M_2/N_2 relationship. All models retain the same segment constants. They do not average amplitudes fitted separately to individual segments.

The amplitude fits use $W_{ii} = 1/\sigma_i^2$, where $\sigma_i = \alpha_{p(i)}/\sqrt{n_i}$, n_i is the retained bin-member count and α_p is the period's one-second frequency-noise coefficient. The coefficients are derived from overlapping Allan deviations at 256–4,096 s within acquisition segments. The reported marginal errors are the larger of the residual-scaled least-squares and UTC-day-deletion jackknife standard errors; each deletion refits the coefficients. The full-template estimate is 0.9551 ± 0.1082, and the grouped lunar and neighbouring estimates are 1.0004 ± 0.1320 and 0.9251 ± 0.3649. In the phase-relaxed fit, the lunar in-phase and quadrature estimates are 1.0007 ± 0.1320 and -0.0969 ± 0.1989. The contours in Fig. 3b instead use the scaled-least-squares covariance at radii one and two; its quadrature marginal standard error is 0.1273. These contours are covariance summaries, not independently calibrated joint confidence regions.

Local spectral density in Figure 3

Figure 3c uses the original fractional-frequency means of the auxiliary reduction, equal row weights, and their stored UTC bin-centre times. The observation vector and each trial sine–cosine pair are projected against the same 54 segment constants. The spectral score $\Delta\mathrm{RSS}(f)$ is the reduction in residual sum of squares obtained by fitting the projected pair. The residual curve is calculated after subtracting the fixed unit-amplitude complete physical prediction; the response coefficients of Fig. 3a are not

applied in this subtraction. With dimensionless fractional frequency y, the density is $S_y(f) = 600\,\Delta\mathrm{RSS}(f)$ per Hz, or $600 \times 10^{-6}\Delta\mathrm{RSS}(f)$ per µHz. The plot uses the nominal-600-s projection normalization on the observed sampling.

The left axis expresses equivalent-height density as $S_{h,\mathrm{cm}} = 10^4 c^4 S_y / g_0^2$ in cm²/µHz, with conventional standard gravity $g_0 = 9.80665\ \mathrm{m/s^2}$ and $c = 299,792,458\ \mathrm{m/s}$. The right axis gives the same density in fractional-frequency units. Both axes are linear and start at zero. This conversion represents the height equivalent of a potential difference and does not alter the physical model's gravity value.

The plotted band is 21–24.5 µHz. The stored scan maxima near M_2 and S_2 occur at 22.361 and 23.147 µHz at the reported precision, distinct from the prescribed astronomical-frequency markers. The ratios $10\log_{10}[S_{y,\mathrm{observed}}(f)/S_{y,\mathrm{residual}}(f)]$ at those stored scan points are 16.7 and 9.3 dB. They compare before-and-after spectra on identical sampling; the residual includes measurement fluctuations and model mismatch.

Auxiliary frequency-domain analysis

The separate non-overlapping reduction contains 3,299 means and 545.924 member-hours in 54 segments. We use its recorded fractional-frequency means, expressed numerically in units of 10^{-18}, at their stored UTC bin centres. The trial sinusoids are evaluated at these centres, following the bin-centre spectral convention; within-bin member integration and the main analysis's triangular filter are not reconstructed here. Equal spectral weights differ from the noise-based weights used for the physical amplitude estimates in that reduction.

For a nuisance matrix N, we form its orthonormal basis Q and apply $R = I - QQ^T$ to both the observation vector and each trial cosine–sine pair. The spectral score ΔRSS is the reduction in residual sum of squares when the projected pair is fitted. We use the nominal-600-s normalization, $D(f) = 600 \times 10^{-42}$ ΔRSS per µHz. This convention provides a common scale for comparisons on these gapped observations; integrating it is not asserted to recover the time-domain variance. Extended Data Fig. 7a uses segment constants only. Extended Data Fig. 7b–d add freely phased N_2 and K_2 terms. All sinusoidal phases refer to 6 September 2025, 16:05 UTC, the first stored bin centre.

The M_2 and S_2 subtraction waveforms come from a simultaneous four-coefficient fit after the same nuisance projection. Conditional scans add the other principal pair to N and scan within ±0.16 µHz of the target. No physical amplitude ratio or phase constraint links the two principal components. As a nuisance sensitivity check, their joint amplitudes without N_2/K_2 control are 4.63 and 1.82×10^{-18}. Scaled least-squares and UTC-day-deletion covariances quantify the joint coefficient uncertainties (Extended Data Table 4 and Extended Data Fig. 9). Synthetic injection curves contain only the fitted principal sinusoidal waveform evaluated at the measured times, with no added random noise. Direct nested regression verifies the conditional ΔRSS calculation.

Compensation sensitivity and template correlation

For donor D and target T, the joint fit supplied the fixed donor coefficient vector $\hat{\beta}_D$. For 61 values of λ from 0 to 1.5, we formed $y_\lambda = y - \lambda X_D \hat{\beta}_D$ and fitted only the target pair together with N. With $P_T = R_N X_T$ and $P_D = R_N X_D$, its coefficient vector obeys

$$\hat{\beta}_T(\lambda) = \hat{\beta}_{T,\text{single}} - \lambda P_T^{+} P_D \hat{\beta}_D.$$

We also repeated the sweep using the donor's single-frequency estimate to show the dependence on waveform definition. A separate jointly controlled fit retains both M_2 and S_2 pairs after subtraction. Its target coefficient is necessarily invariant, because the subtracted waveform lies within the fitted donor subspace; this curve is an algebraic control, not evidence against leakage. Amplitude is the norm of the two coefficients and phase is $\text{atan2}(\beta_{\sin}, \beta_{\cos})$, corresponding to $A\cos(2\pi f(t - t_0) - \phi)$. Target-only incremental improvement is $\Delta\text{RSS} = \hat{\beta}_T^{\mathsf{T}} P_T^{\mathsf{T}} P_T \hat{\beta}_T$. The jointly controlled improvement instead residualizes the target against both N and the donor, so these two improvements have different baselines even where their coefficient estimates coincide. Density displays use the nominal 600-s least-squares convention $600\Delta\text{RSS} \times 10^{-42}$ per µHz for observations expressed in units of 10^{-18}; an integral of this gapped-record display is not asserted to satisfy a Parseval variance identity.

Canonical correlations are the singular values of $Q_M^{\mathsf{T}} Q_S$, where each Q spans a residualized constituent pair. Template lag correlations evaluate the fitted donor waveform at $t - \ell$ and the target at t, residualize both on the unchanged observation support, and normalize their inner product. Lags span ±24 h in 5-min steps. Their zero-lag correlation is 0.1955; maximum absolute lag correlations are 0.2413 for shifted M_2 against S_2 and 0.2113 in the reverse direction. These are design and fitted-template similarities, not an observed stochastic cross-correlation function or a measured propagation delay.

For alias diagnostics, we scanned 20–25 µHz with spacing $1/(12T)$, where T is the elapsed observation span. For each donor, we selected the three strongest local maxima of its pure-injection spectrum at least $2/T$ away from its own frequency. Selection used the injection response, not a significance threshold on the observed data. At each selected frequency, we compared the actual refit to the predicted vector response $\hat{b}_f(\lambda) = \hat{b}_f(0) - \lambda \hat{b}_{f,D}$. Its squared norm in the frequency-specific Gram metric gives the predicted quadratic power response. Numerical agreement verifies the processing operator; because this is a linear identity, agreement alone cannot establish an exclusively aliased origin of the observed peak. The compensation curves describe deterministic sensitivity conditional on the fitted waveforms.

We express each four-coefficient covariance as a correlation matrix, $R_{ij} = C_{ij}/\sqrt{C_{ii}C_{jj}}$, separately for scaled least squares and UTC-day deletion (Extended Data Fig. 9c,d). These matrices describe the coupled uncertainty of the joint coefficient estimates, distinct from correlations of the design columns and fitted templates.

Author contributions

Jian-Wei Pan planned and directed the study and contributed to manuscript writing. Hai-Feng Jiang designed and coordinated the optical-clock comparison and organized its implementation. Qiang Zhang designed and coordinated the comparison experiment. Lei Hou coordinated the remote frequency comparison and was responsible for the optical frequency combs at both stations. Xiang-Pei Liu designed and coordinated the Sr-clock comparison and evaluated its gravitational redshift. Qi Shen designed the optical-frequency-comb and clock-comparison experimental schemes. Sheng-Kai Liao, Juan Yin and Cheng-Zhi Peng contributed to discussions of the experimental design and planning.

Han-Ning Dai was responsible for the overall Sr-clock design. Xiang Zhang operated and maintained Sr1 and analysed the Sr-clock data. Jia-Hao Zhang and Long-Fei Liao operated and maintained the Sr clock. Xiao Jiang and Yu-Ao Chen were responsible for the Sr-clock system. Yu-Chen Zhang was responsible for the link between the Sr1 ultrastable laser and the frequency comb and for height measurements in Shanghai. De-Quan Kong maintained the U1 ultrastable laser. Xing-Yang Cui constructed and maintained the Sr-clock laser servo systems; Ping Xu constructed and maintained its ultrastable clock-laser system; Peng Liu constructed and maintained the atomic-physics system; and Jie Li constructed and maintained its electronics. Hua-Qing Zhang optimized and modified the Sr-clock laser output.

Zhuan-Xian Xiong and Baolong Lyu were responsible for the overall Yb-clock system. Qiang Zhu constructed the Yb-clock optical system, designed its electronics and analysed data. Bing Wang constructed the optical system and operated and maintained the Yb clock; Ling-Xiang He operated and maintained the Yb clock. De-Zhi Xiong performed simulations and analysis of the Yb-clock blackbody-radiation shift and ultrastable-cavity vibrations. Baolong Lyu also performed theoretical analysis and data processing for the AC Stark-shift uncertainty.

Fa-Xi Chen was responsible for the fibre link, and Li-Bo Li maintained it. Wen-Xuan Song operated the Shanghai optical frequency comb and acquired and processed data. Jian-Yu Guan was responsible for the optical frequency combs and comparison-data acquisition and processing. Shao-Mao Wang was responsible for the 1,550-nm ultrastable laser used for frequency transfer in the local frequency-comb system. Youchao Xie, Lei Wang and Wen-Bin Shen evaluated the gravitational potentials in Wuhan and Shanghai. Yu-Ao Chen also processed data.

Lei Hou, Wen-Xuan Song, Jian-Yu Guan, Han-Ning Dai, Qiang Zhang, Yu-Ao Chen, Hai-Feng Jiang and Jian-Wei Pan contributed to manuscript writing.

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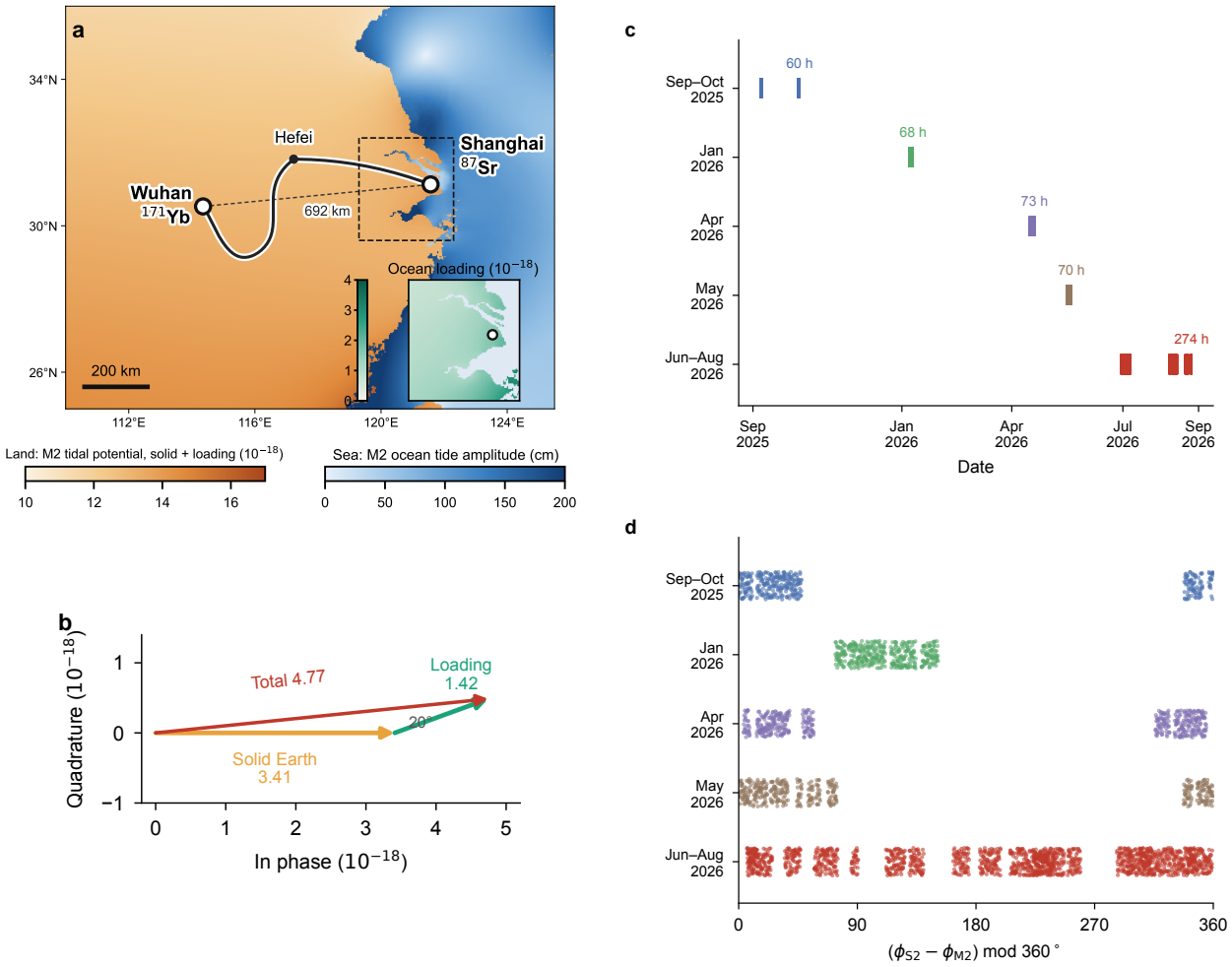


Fig. 1 | The predicted tide on a coastal-inland baseline. **a,** The Sr clock in Shanghai, the Yb clock in Wuhan and the fibre link through Hefei. The dashed line marks the 692-km distance between the clocks. Land colours show the predicted M_2 tidal potential sensed by a clock, including the solid Earth tide and ocean loading, expressed as fractional frequency. The inset shows the ocean-loading contribution around Shanghai; sea colours show the FES2022 M_2 ocean-tide amplitude. **b,** Predicted differential M_2 phasors of the solid Earth tide, the ocean-loading potential and their sum, referenced to the solid Earth tide. **c,** Filtered samples and observation hours in the five campaigns from September 2025 to August 2026. **d,** Relative astronomical phase of S_2 and M_2, including the nodal correction, at the same samples. The short campaigns cover limited relative phases, whereas the summer campaign covers the full cycle as the lunar tide moves through the solar day.

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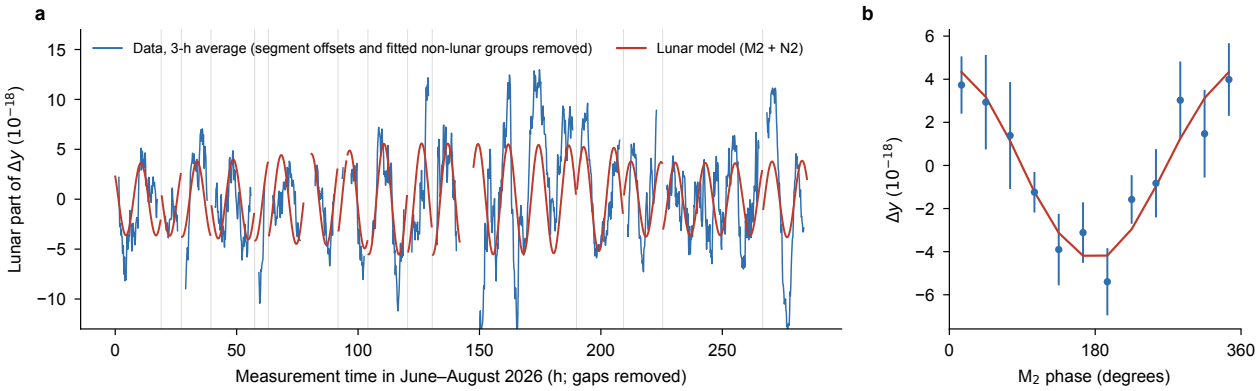


Fig. 2 | The lunar tide in the summer record. a, Three-hour averages of the June–August 2026 frequency ratio after subtracting fitted segment offsets and the fitted near-12-h and diurnal tidal groups (blue), with the lunar model (red). The measurement-time axis compresses gaps longer than 6 h; grey lines mark their removal. White frequency noise alone corresponds to a scatter of approximately 5×10^{-18} for a 3-h average. **b,** The same data folded at the model M_2 phase in 30° bins, with the lunar model (M_2 and N_2) folded using identical sampling. Error bars are day-by-day jackknife uncertainties conditional on the fitted non-lunar terms, which are held fixed during resampling. No empirical daily terms are fitted. Extended Data Fig. 4 provides a complementary view with the actual calendar-time axis.

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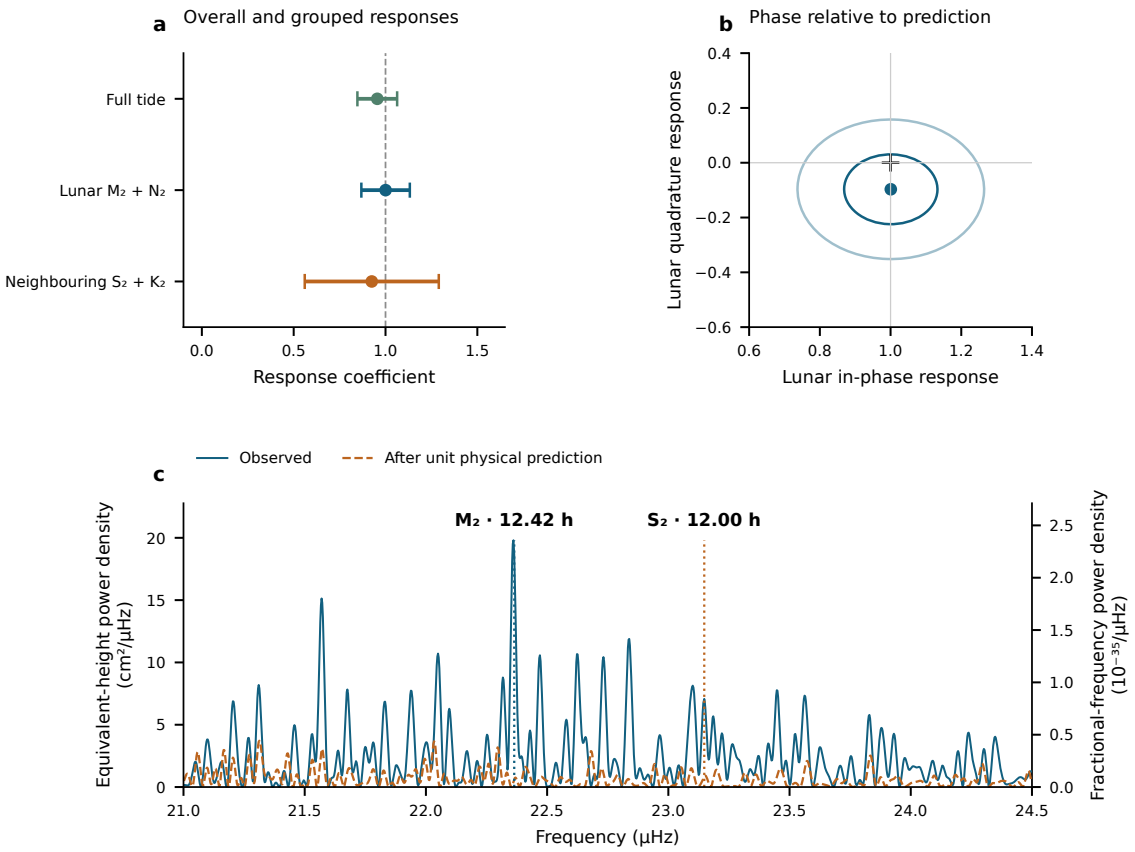


Fig. 3 | Amplitude, phase and frequency of the tidal response. All panels use the auxiliary reduction into 3,299 non-overlapping 600-s means, retaining 54 segment constants. **a,** Complete-template response and jointly fitted lunar (M_2, N_2) and neighbouring semidiurnal (S_2, K_2) responses, using the period-dependent noise weights. Error bars are one statistical standard error, the larger of scaled least squares and UTC-day-deletion jackknife errors. The complete-template and grouped coefficients come from separate fits to the same observations. **b,** Lunar in-phase and quadrature coefficients; the cross marks $(1, 0)$. Contours have radii one and two in the scaled-least-squares covariance. The marginal working errors are 0.13 and 0.20; the scaled-least-squares quadrature error is 0.127. **c,** Observed spectrum over 21–24.5 μHz and the spectrum after subtracting the fixed unit-amplitude physical prediction. The scan uses equal weights and projects both data and trial sinusoids against the segment constants. Both vertical axes are linear and start at zero. The equivalent-height density on the left, in cm²/μHz, and fractional-frequency density on the right obey $S_{h,\mathrm{cm}} = 10^4 c^4 S_y / g_0^2$, with $g_0 = 9.80665\ \mathrm{m/s^2}$. Markers indicate the prescribed M_2 and S_2 frequencies. Sampling and conditional scans are shown in Extended Data Fig. 7. The primary triangular-filter results are presented with their methods in Extended Data Fig. 11.

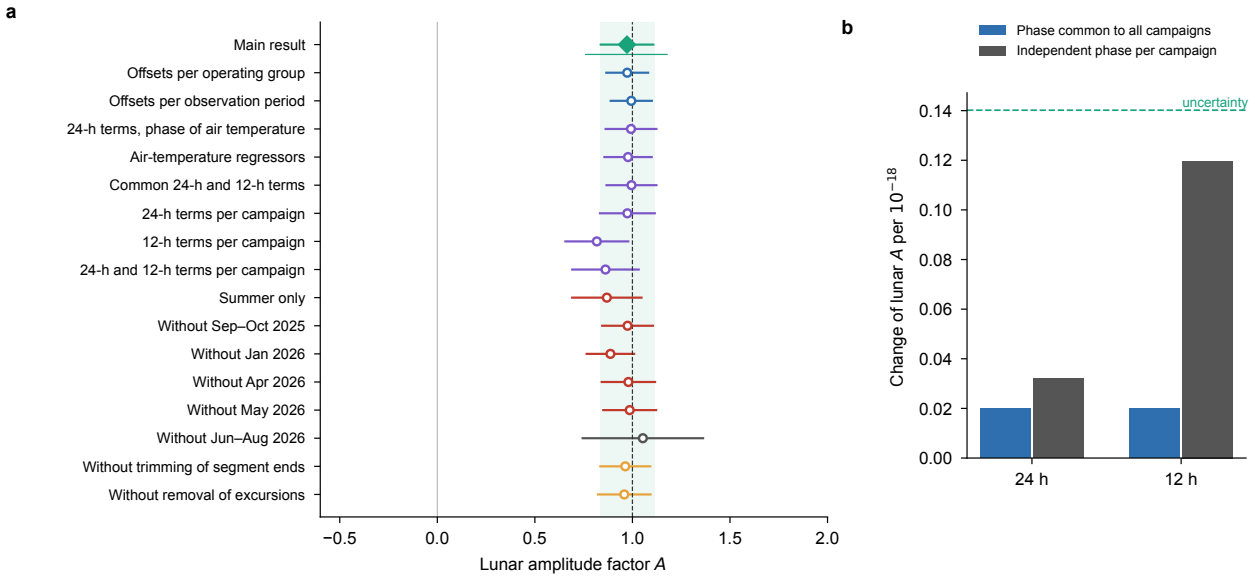
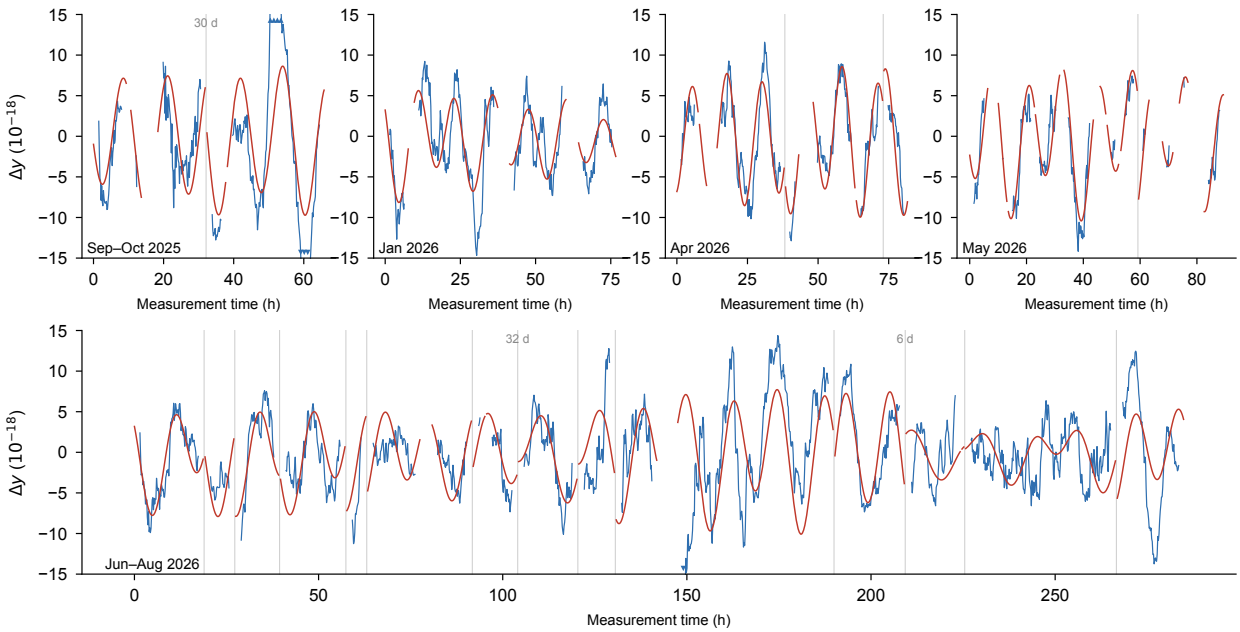


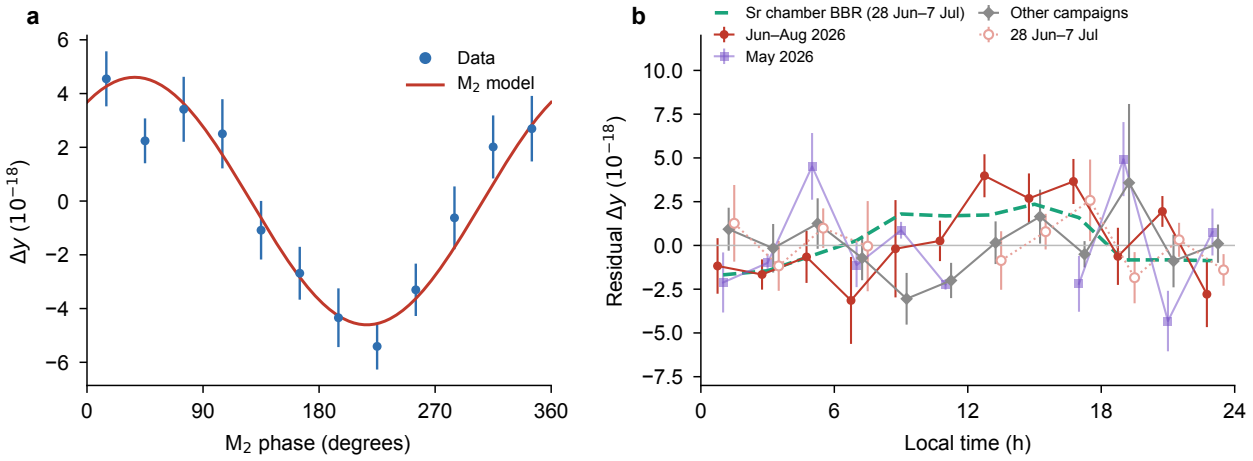
Fig. 4 | Robustness of the lunar amplitude. **a,** Lunar amplitude for the main fit (filled diamond and shaded uncertainty band), alternative offset definitions, daily terms represented by air-temperature measurements or fitted 24-h and 12-h sinusoids, the summer campaign alone, omission of each campaign, and retention of segment ends or transient excursions. The thin bar below the main result is the 95% interval from resampling operating groups. These estimates share observations. The grey interval obtained without the summer campaign shows the loss of information that separates the tidal groups. **b,** Change in the lunar amplitude caused by a 24-h or 12-h sinusoid of fractional-frequency amplitude 10^{-18}, either with one phase common to all campaigns or with the combination of independent campaign phases that produces the largest change. The dashed line marks the uncertainty of the main estimate. The vertical axis is linear.

Table 1 | Related tidal estimates. These estimates share observations and are not combined. The last column specifies each uncertainty definition. Applying the segment-wise estimator of Hou et al.[17] to the summer record with the tidal model used here gives 0.78 ± 0.19.

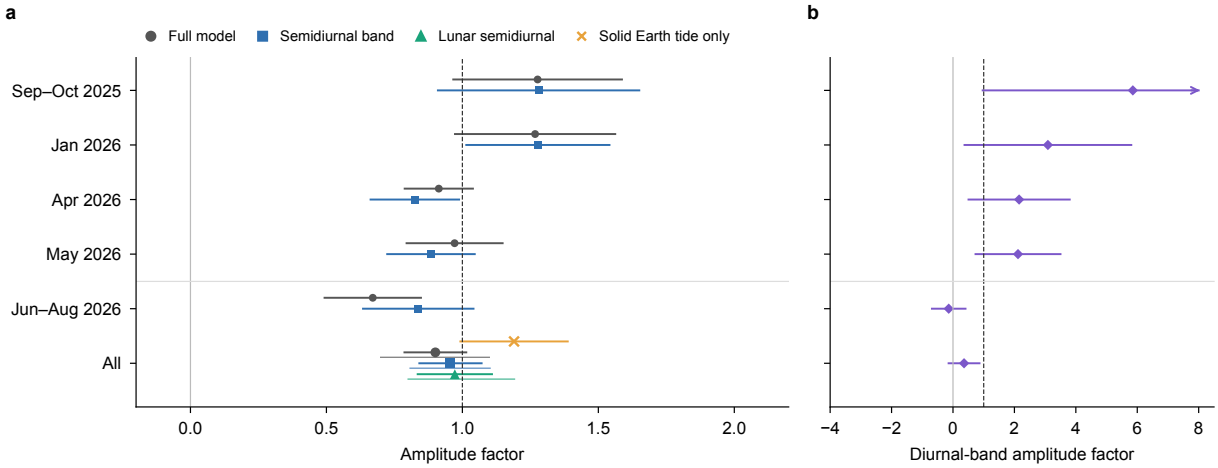
Analysis	Data and model	Amplitude factor A	Uncertainty
Hou et al.[17]	Summer; 16 segments; their tidal model and segment-wise correlation	0.87(15)	Reported weighted statistical error
Main analysis	54 segments; lunar group; segment offsets; astronomical ephemerides	0.97 ± 0.14	Calibrated by residual resampling
Full-template analysis	Same observations; full template; segment offsets	0.90 ± 0.12	Calibrated by residual resampling
Alternative processing	2026 observations; 600-s means; ten-day calendar baselines; full template	0.88 ± 0.10	Largest of least-squares, delete-day and delete-period standard errors
Alternative processing	Same observations and baselines; lunar group; daily terms in each ten-day block	0.89 ± 0.28	Largest of least-squares, delete-day and delete-period standard errors



Extended Data Fig. 1 | Tidal records in all campaigns. Three-hour averages of the filtered frequency ratio after subtraction of fitted segment offsets (blue), with the tidal model (red). Measurement-time axes compress gaps longer than 6 h, marked by vertical lines; gaps longer than 5 days are labelled. The vertical range is ±15 × 10^{-18}, and triangles mark values outside this range. The actual calendar-time axis is retained for the April campaign in Extended Data Fig. 4a.

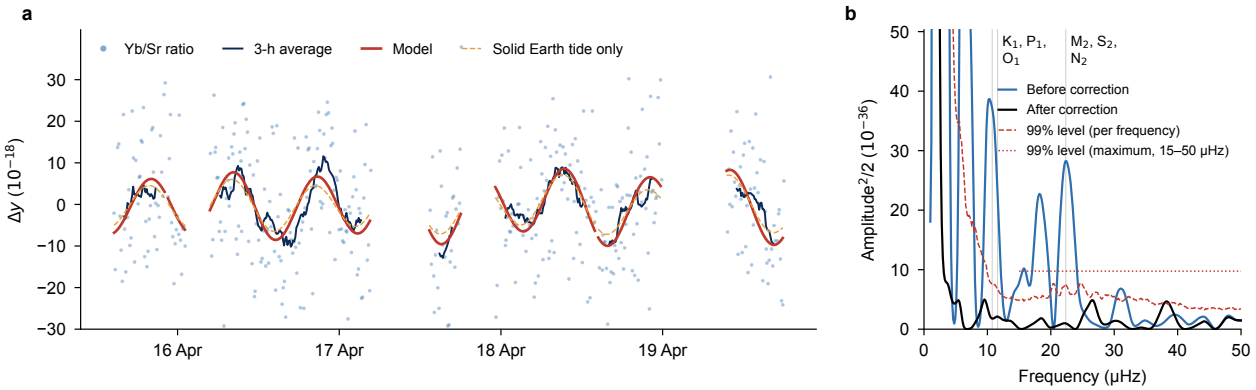


Extended Data Fig. 2 | Folds by tidal phase and local time. a, All observations folded at the M_2 phase after subtracting fitted segment offsets and all tidal constituents except M_2 from the full-model fit. Points are means in 30° bins with day-by-day jackknife uncertainties; the curve is the M_2 model. These phase-bin uncertainties are conditional on the fitted terms removed before folding, which are held fixed. **b,** Tidally corrected residuals folded by local time in 2-h bins for summer, May (light symbols), the other three campaigns, and summer observations overlapping the temperature record from 28 June to 7 July (open circles). Error bars are day-by-day jackknife uncertainties; bins covering fewer than three days are omitted. The dashed curve is the mean daily blackbody-radiation shift computed from the Sr chamber temperature from 28 June to 7 July 2026 after removing a 24-h running mean and excluding the temperature transient on 30 June. The real-time correction removes this chamber-temperature shift from the recorded frequency ratio.

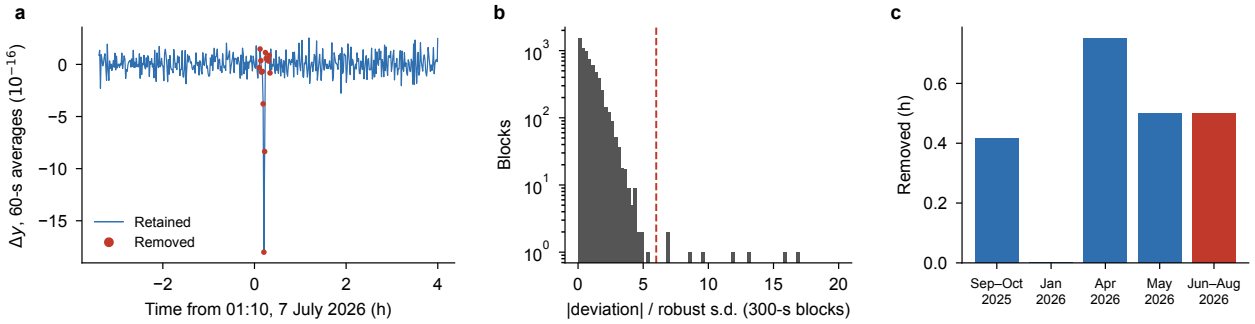


Extended Data Fig. 3 | Campaign-resolved tidal amplitudes. a, Full-model and semidiurnal amplitude factors for each campaign and the combined observations, fitted with segment offsets. Combined-data estimates also include the lunar group and the solid Earth tide alone. Short campaigns constrain the semidiurnal band as a whole; the joint fit separates the lunar group. **b,** Diurnal amplitude fitted jointly with the semidiurnal band in a two-band fit; the combined estimate is 0.36 ± 0.50. The arrow marks an uncertainty extending beyond the plotted range. Error bars represent one standard deviation, calibrated by residual resampling for the full model, semidiurnal band and lunar group; for the diurnal band and solid Earth tide alone they are the larger of the scaled least-squares and day-by-day jackknife uncertainties. Thin lines in the combined-data rows show 95% intervals from resampling operating groups, or segments for the lunar group.

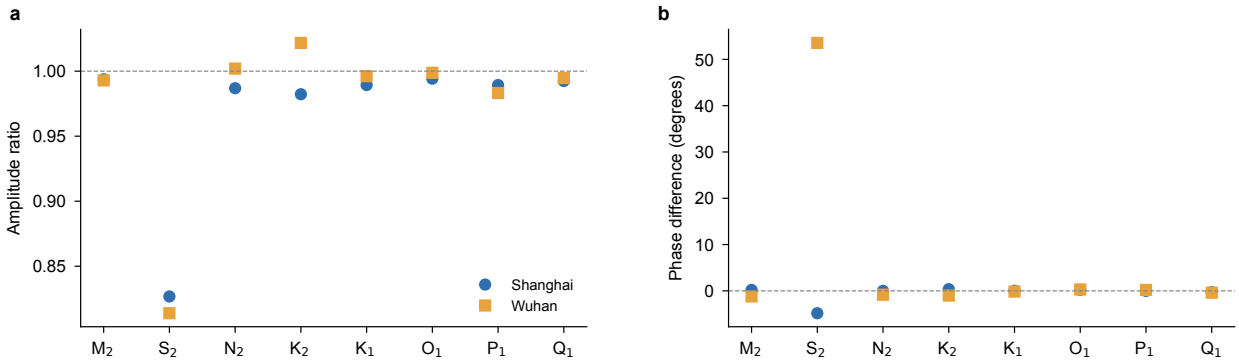
760



Extended Data Fig. 4 | April campaign in calendar time and frequency. a, Frequency-ratio observations after the 1,200-s triangular filter and removal of fitted segment offsets, their 3-h averages, and predictions with and without ocean loading. The horizontal axis retains actual calendar dates and gaps (local time, UTC+8). **b,** Sinusoidal power before and after subtracting the model, expressed as amplitude squared divided by two on a linear vertical axis, in units of 10^{-36}. The dashed curve is the 99% level at each frequency; the dotted line is the 99% level for the maximum over 15–50 μHz, both obtained from 1,000 resampled residual series. The four-day record combines K_1, P_1 and O_1 into a diurnal band and M_2, S_2 and N_2 into a semidiurnal band. Power below 8 μHz exceeds the displayed range.

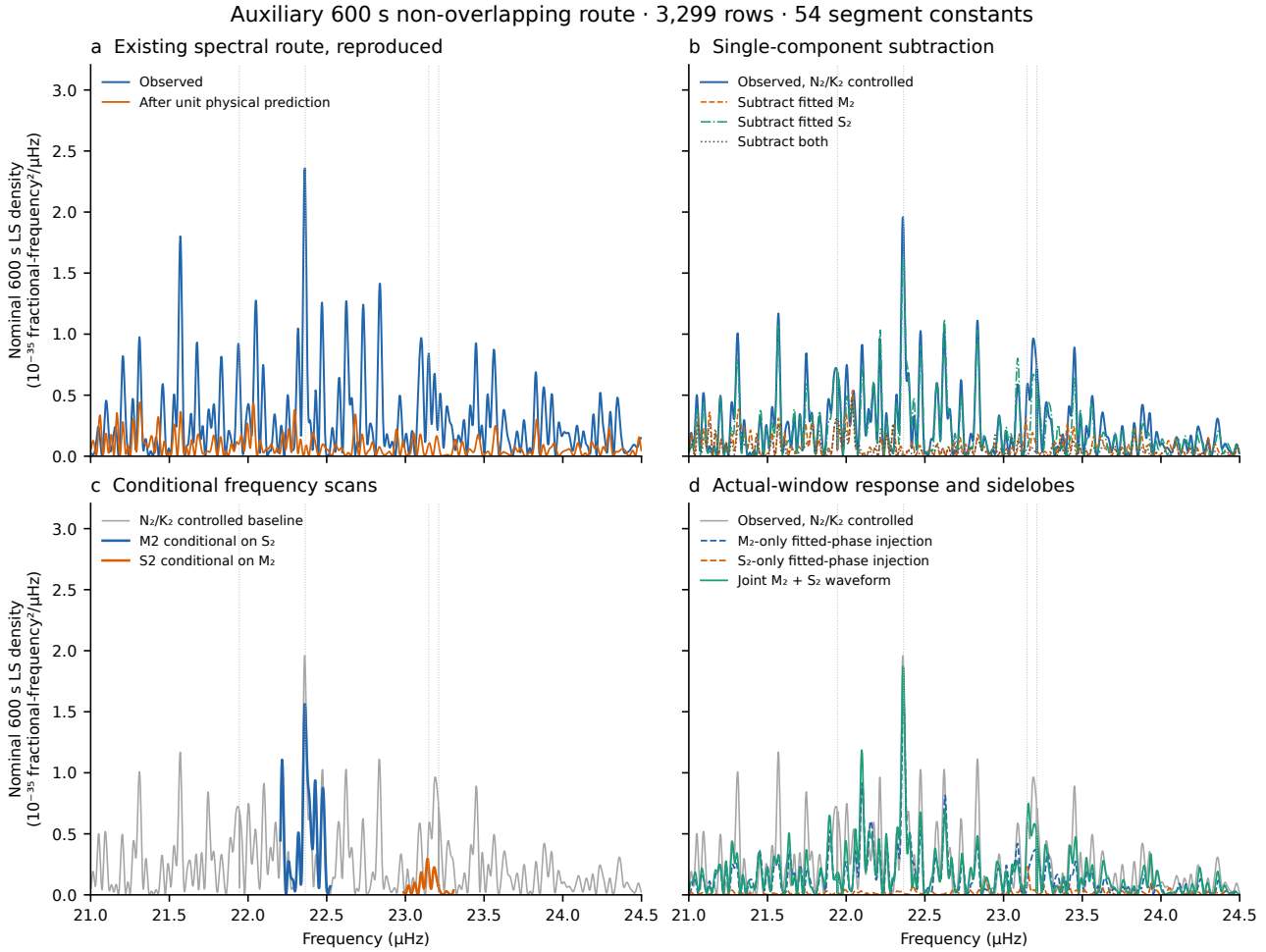


Extended Data Fig. 5 | Transient excursions and sample exclusion. **a,** Sixty-second averages around the excursion on 7 July 2026; red points indicate the 1-s samples excluded by the criterion in Methods. **b,** Distribution of deviations of 300-s means from a running median over ±30 min, divided by the robust standard deviation within each segment. The dashed line marks the threshold of six. The count axis is logarithmic. **c,** Observation time excluded in each campaign.

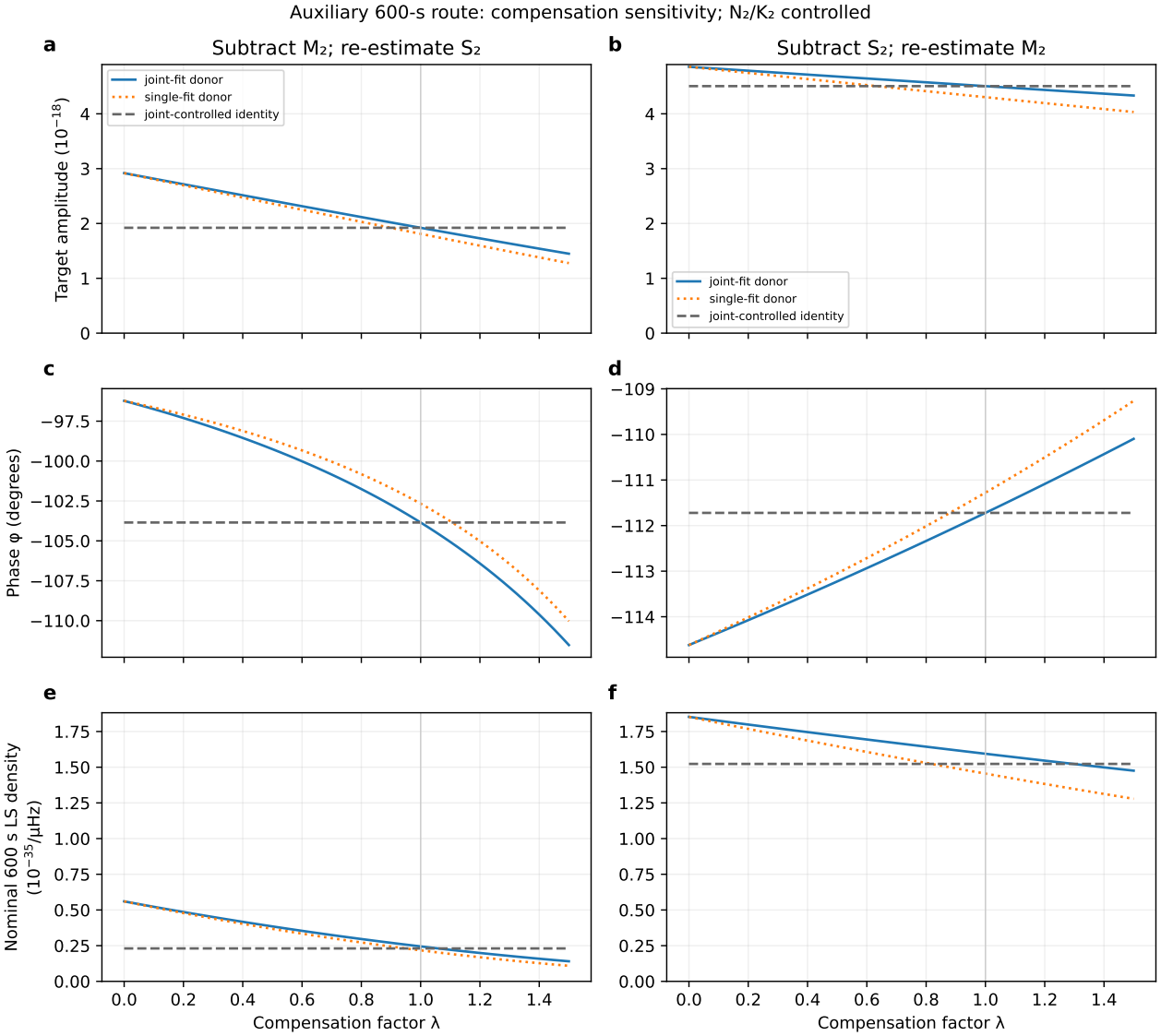


Extended Data Fig. 6 | Radial-displacement validation of the loading convolution. a,b, Amplitude ratios (**a**) and phase differences (**b**) between radial displacements calculated with the loading Green's function and the FES2022 load-tide grid in the centre-of-mass frame, at Shanghai and Wuhan, for eight constituents. Methods discusses the S_2 discrepancy.

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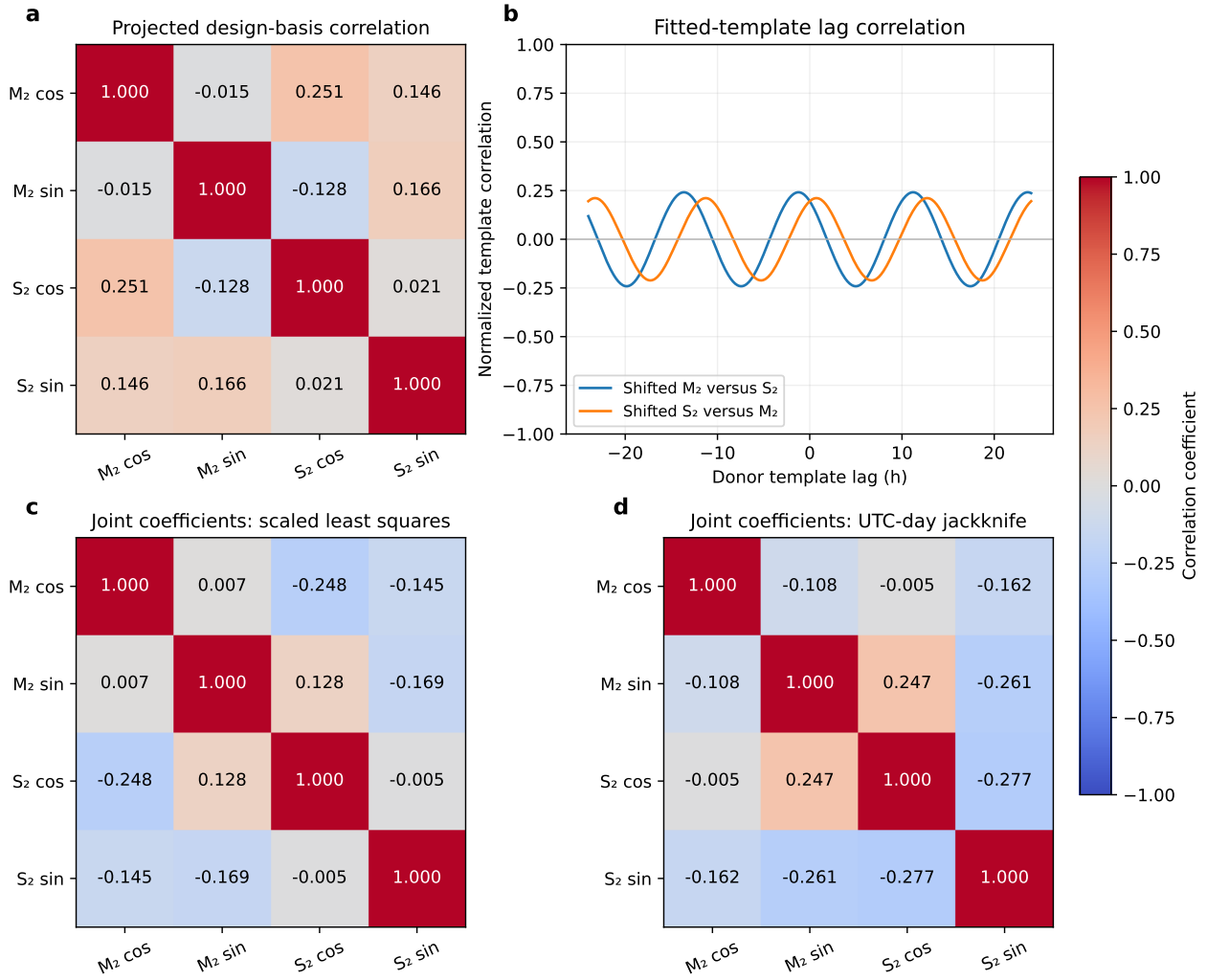


Extended Data Fig. 7 | Local spectra and sampling-response checks on the auxiliary 600-s non-overlapping reduction. All panels retain 3,299 recorded bin-centre times and the gaps between them, use equal weights and 54 segment constants, and share a linear ordinate and identical density normalization. **a,** The existing observed spectrum, reproduced from the frozen observations, and the stored spectrum after subtraction of the fixed unit physical prediction. **b,** Spectra before subtraction, after subtracting the fitted M_2 waveform, after subtracting the fitted S_2 waveform, and after subtracting both. The two waveforms are obtained from one simultaneous freely phased fit with N_2 and K_2 controlled by their own sine–cosine pairs. **c,** Local scans around M_2 conditioned on S_2, and around S_2 conditioned on M_2; N_2 and K_2 remain controlled. Both the response and the scanned sine–cosine pair undergo the same nuisance projection. The grey curve is the baseline from b. **d,** Deterministic M_2-only, S_2-only and combined fitted-waveform injections through the same sampling and projection operator, compared with the observed spectrum. Individual injected powers need not sum to the combined power because of interference. Dotted vertical lines mark N_2, M_2, S_2 and K_2, in ascending frequency. The ordinate is the nominal-600-s least-squares density defined in Methods, not the amplitude²/2 power in the main triangular-filter analysis. No observations are added in gaps, and these auxiliary estimates do not replace the main lunar-group response.

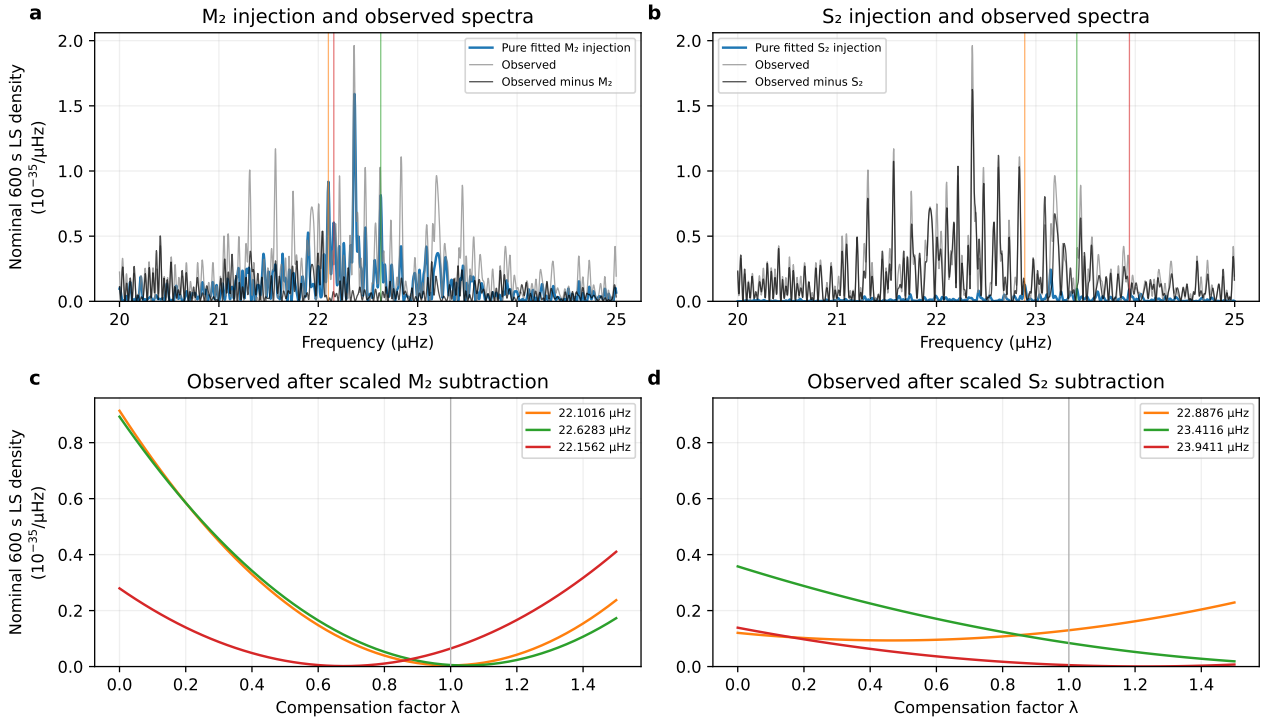


Extended Data Fig. 8 | Compensation sensitivity on the auxiliary 600-s record. a,c,e, Subtracting a scaled M_2 waveform and re-estimating S_2; b,d,f, the reverse. a,b, Target amplitude; c,d, absolute phase; e,f, nominal 600-s least-squares density. Solid curves use donor waveforms from the joint fit; dotted curves use single-frequency donor fits. All target-only fits retain segment offsets and free N_2/K_2 pairs. Dashed grey curves also control the donor pair and are invariant by construction. The power improvement for these curves is conditional on the donor and therefore has a different baseline from the target-only curves. $\lambda = 1$ denotes subtraction of the selected estimated waveform. Curves describe deterministic sensitivity conditional on the fitted waveforms. Amplitude and density panels share their respective vertical scales across directions, and density axes are linear with zero baselines.

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Extended Data Fig. 9 | Harmonic geometry and joint-coefficient correlations on the auxiliary 600-s record. a, Correlations of the M_2 and S_2 cosine and sine design columns after projecting out 54 segment constants and free N_2/K_2 pairs. b, Normalized lag correlations of the fitted templates, evaluated on the original timestamps without interpolating missing observations; these describe template similarity, not propagation delay. c,d, Correlations between the four coefficients estimated jointly, obtained by normalizing the existing scaled least-squares covariance (c) and UTC-day delete-one jackknife covariance (d). Coefficient order is M_2 cosine, M_2 sine, S_2 cosine and S_2 sine. The three matrices use the same -1 to $+1$ colour scale. Joint-coefficient covariance describes uncertainty coupling and is distinct from the design-column correlations in a. The 43 UTC days define deletion blocks for the jackknife, not 43 independent measurements. The joint correlations complement the conditional and subtraction evidence for the detected frequency components.



Extended Data Fig. 10 | Sampling aliases and compensation response. a,b, Observed spectra, spectra after subtraction of a joint-fit donor, and spectra of that donor alone on the identical timestamps and nuisance projection. Vertical markers identify three off-centre local maxima selected from the injection. c,d, Observed power at those fixed candidate frequencies as the compensation factor varies. The curves equal the quadratic responses predicted from the complex coefficient subtraction; this agreement is an operator check rather than an independent alias attribution. All density axes are linear and share scales within rows.

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Extended Data Table 1 | Noise-model simulations of the alternative processing. Bias and coverage of nominal 95% intervals based on the largest of the least-squares, delete-day and delete-period standard errors. T denotes the full template without daily terms, and L the lunar group with daily terms, using the ten-day calendar-baseline design in Table 1 and the actual observation times. Gaussian scenarios contain 1,024 simulations; Student-t and moving-average scenarios contain 4,096. OU denotes Ornstein–Uhlenbeck noise and MA(1) a first-order moving-average process.

Noise model	Bias (T)	Coverage (T), %	Bias (L)	Coverage (L), %
White	−0.000	96.7	−0.001	98.3
White + OU, 1 h	−0.007	94.5	−0.002	96.9
White + OU, 12 h	−0.002	96.1	+0.031	98.8
White + OU, 3 d	+0.003	95.9	+0.004	99.8
White + period constants	+0.001	96.0	+0.004	98.3
Student-t_7 MA(1)	+0.002	98.3	+0.004	98.2
Student-t_5 MA(1)	+0.001	98.9	+0.003	98.3
Student-t_7 MA(1) + OU	+0.001	97.2	+0.005	98.5

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Extended Data Table 2 | Injection tests of the main estimator. Lunar amplitudes fitted with segment offsets and no empirical daily terms to resampled residuals plus either the full tidal model (Atrue = 1) or the model without the lunar group (Atrue = 0). The four daily-disturbance scenarios are defined in Methods, with 300 realizations per row. Bias and standard deviation describe the fitted amplitude. The median standard error is that computed within each realization, before the final residual-based calibration. Coverage is the fraction of intervals containing the injected amplitude; the accompanying 95% binomial interval refers to one-standard-error coverage. The last column gives the fraction with $A > 3\sigma$.

Disturbance	Atrue	Bias	s.d.	Median error	1σ coverage	95% binomial interval	1.96σ coverage	$A > 3\sigma$
None	0	−0.026	0.134	0.135	0.70	0.645–0.751	0.94	0.003
None	1	−0.003	0.135	0.137	0.71	0.659–0.764	0.94	1.000
24 h, air-temperature phase	0	−0.013	0.141	0.136	0.61	0.549–0.662	0.93	0.000
24 h, air-temperature phase	1	−0.007	0.145	0.138	0.64	0.579–0.691	0.94	1.000
Common 24 h + 12 h	0	−0.013	0.148	0.140	0.63	0.573–0.685	0.90	0.000
Common 24 h + 12 h	1	−0.024	0.148	0.139	0.60	0.546–0.659	0.92	1.000
Summer 24 h + 12 h	0	−0.021	0.138	0.140	0.67	0.610–0.720	0.95	0.000
Summer 24 h + 12 h	1	−0.016	0.150	0.137	0.59	0.529–0.643	0.92	1.000

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Extended Data Table 3 | Daily terms and campaign-specific tidal amplitudes. Amplitude and local time of maximum of the 24-h term, and amplitude of the 12-h term, fitted jointly with the full tidal model within each campaign. Daily amplitudes are expressed in 10^{-18}. Their uncertainties are the larger of the scaled least-squares and day-by-day jackknife values; uncertainties in the time of maximum are jackknife estimates. Weak sinusoidal amplitudes are biased positive. The full-template amplitude A, fitted without daily terms, and its residual-calibrated uncertainty are those in Extended Data Fig. 3a. The last column gives the lunar amplitude fitted independently to each campaign with the main estimator.

Campaign	Days	24-h amplitude	24-h maximum, h	12-h amplitude	A	Alunar
Sep–Oct 2025	4	5.1 ± 2.7	20.7 ± 5.6	5.0 ± 9.7	1.28 ± 0.31	0.04 ± 0.74
Jan 2026	4	2.1 ± 3.0	17.2 ± 3.9	4.1 ± 1.8	1.27 ± 0.30	1.36 ± 0.37
Apr 2026	5	2.9 ± 2.2	1.9 ± 1.4	1.0 ± 3.9	0.91 ± 0.13	0.66 ± 0.42
May 2026	4	5.2 ± 3.3	3.8 ± 2.4	3.1 ± 2.6	0.97 ± 0.18	0.28 ± 0.68
Jun–Aug 2026	22	2.8 ± 1.3	15.8 ± 1.5	0.6 ± 0.7	0.67 ± 0.18	0.87 ± 0.18

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Extended Data Table 4 | Free sinusoidal coefficients in the auxiliary reduction. Joint M_2 and S_2 cosine–sine coefficients, after controlling 54 segment constants and freely phased N_2 and K_2 terms. Coefficients and standard errors are in units of 10^{-18}. Phases refer to 6 September 2025, 16:05 UTC. Scaled least-squares errors use the residual variance; UTC-day jackknife errors delete each of the 43 observed dates and refit all coefficients. The full covariance matrices are retained with the analysis. These errors describe this auxiliary harmonic model and are not substituted for the primary lunar-group uncertainty.

Coefficient	Estimate	Scaled LS standard error	UTC-day jackknife standard error
M_2 cosine	-1.667	0.688	0.880
M_2 sine	-4.185	0.679	0.635
S_2 cosine	-0.460	0.738	0.701
S_2 sine	-1.866	0.750	1.043